

Total No. of Questions : 8]

SEAT No. :

PC-2497

[Total No. of Pages : 3

[6354]-626

B.E. (Mechanical)

ENERGY ENGINEERING

(2019 Pattern) (Semester - VIII) (402049)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data wherever necessary and mention the same clearly.
- 5) Use of steam tables, Mollier chart and calculator is allowed.

- Q1) a) Discuss the main factors considered for site selection of Diesel Power Plants. Also Elaborate the advantages and limitations of Diesel Power Plants. [8]
- b) Draw the hydrograph and flow duration curve for the following data: [9]

Month	Inflow(m ³ /s)	Month	Inflow(m ³ /s)
Jan.	1600	July	3000
Feb.	1200	Aug.	3000
March	800	Sept.	1600
April	800	Oct.	800
May	800	Nov.	800
June	1200	Dec.	1000

Determine :

- i) Storage capacity for a constant demand of 1100 m³/s.
- ii) Number of additional months this storage capacity can be utilized if there is no rain fall.

OR

P.T.O.

Q2) a) What do you mean by nuclear fission? Discuss in brief. Elaborate functions of following elements of nuclear reactor [8]

- i) Moderators
- ii) Control Rods

b) Describe the Boiling Water Reactor (BWR) with following points. [9]

- i) Labelled Diagram
- ii) Construction and working
- iii) Material for different elements of reactor
- iv) Limitations

Q3) a) Discuss concept of thermal efficiency and work ratio of Gas Turbine cycle. How Intercooling technique helps to improve thermal efficiency of Gas Turbine cycle? Elaborate with neat schematic and T-s diagram. [9]

b) The air enters the compressor of a gas-turbine power plant at 1 bar, 30 degrees Celsius and 162 tons per hour. The maximum cycle temperature, pressure is 650 degrees Celsius, 5 bar respectively. The two stage expansion with reheating pressure of 2.24 bar is used in the plant. In the reheater gas is heated up to maximum cycle temperature. The isentropic efficiency of compressor, first turbine, and second turbine is 80%, 85%, 90% respectively. Take adiabatic index for air gas as 1.4, 1.33 respectively. Take specific heat for air, gas as 1 KJ/Kg-K, 1.15 KJ/Kg-K respectively. Neglect mass flow rate of fuel. Draw cycle arrangement and T-s diagram and determine [9]

- i) The thermal efficiency of cycle
- ii) Power output of plant in MW

OR

Q4) a) What do you mean by cogeneration? Why Cogeneration is needed in gas power cycle? Explain Cogeneration in gas power cycle with simple block diagram and state its advantages. [9]

b) Discuss the Integrated Gasification Combined Cycle (IGCC) plant with cycle arrangement, T-s diagram, merits and demerits. [9]

Q5) a) State the main functions of the circuit breaker. What are different types of the same? Discuss working of any one circuit breaker with a neat sketch. [9]

b) What are different methods thermal energy storage? Explain anyone with simple diagram. Discuss the principles of energy management with storage systems. [8]

OR

- Q6) a)** The equipment in a power station costs Rs. 15,60,000 and has a salvage value of Rs. 60,000 at the end of 25 years. Determine the depreciated value of the equipment at the end of 20 years on the following methods:[9]
- i) Straight line method
 - ii) Diminishing value method
 - iii) Sinking fund method at 5% compound interest annually
- b)** Define and write the equation for the following : [8]
- i) Diversity factor
 - ii) Demand factor
 - iii) Plant capacity factor
 - iv) Plant utilization factor
- Q7) a)** Discuss Low temperature flat plate collector solar power plant with a labelled diagram, working fluid and material of component, merits and demerits. [9]
- b)** Elaborate superheated steam geothermal energy based system with following points [9]
- i) Labelled Diagram
 - ii) Construction and working
 - iii) Advantages
 - iv) Limitations
- OR
- Q8) a)** Discuss Anderson Ocean Thermal Energy system with following points [9]
- i) Labelled Diagram
 - ii) Construction and working
 - iii) Advantages
 - iv) Limitations
- b)** Elaborate working principle of following with simple diagram. [9]
- i) Biomass gasifier
 - ii) Fuel cell

